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COMMON QUARTERLY EXAMINATION SEPTEMBER - 2025

12 TH STD Reg No

TIME 3 Hours **PART III MATHEMATICS** Total Marks : 90

Instructions: 1. Check the question paper for fairness of printing. If there is any lack of fairness, inform the hall supervisor immediately

2. Use Blue or black ink to write and underline. Pencil to draw diagram

PART I

I. Answer all the questions

II. Choose the most appropriate answer from the given four alternatives and write the option code with the corresponding answer **20x1=20**

1) If $|\text{adj (adj } A)| = |A|^9$, then the order of the square matrix A is
(1) 3 (2) 4 (3) 2 (4) 5

2) If $A^T A^{-1}$ is symmetric, then $A^2 =$
(1) A^{-1} (2) $(A^T)^2$ (3) A^T (4) $(A^{-1})^2$

3) If $A = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$ then $|\text{adj } (AB)| =$
(1) -40 (2) -80 (3) -60 (4) -20

4) If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ then rank of $A^T A$ is (1) 3 (2) 0 (3) 1 (4) 2

5) If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^2 + \beta^2$ is
(1) -2 (2) -1 (3) 1 (4) 2

6) The solution of the equation $|z| - z = 1 + 2i$ is
(1) $\frac{3}{2} - 2i$ (2) $-\frac{3}{2} + 2i$ (3) $2 - \frac{3}{2}i$ (4) $2 + \frac{3}{2}i$

7) If $(1+i)(1+2i)(1+3i)\cdots(1+ni) = x + iy$, then $2 \cdot 5 \cdot 10 \cdots (1+n^2)$ is
(1) 1 (2) i (3) $x^2 + y^2$ (4) $1 + n^2$

8) If α, β and γ are the roots of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is
(1) $-\frac{q}{r}$ (2) $-\frac{p}{r}$ (3) $\frac{q}{r}$ (4) $-\frac{q}{p}$

9) The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is
(1) 2 (2) 4 (3) 1 (4) ∞

10) If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{3}$; then $\cos^{-1} x + \cos^{-1} y$ is equal to
(1) $\frac{2\pi}{3}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{6}$ (4) π

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11) If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, the value of $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101}+y^{101}+z^{101}}$ is

(1) 0 (2) 1 (3) 2 (4) 3

12) $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to

(1) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ (2) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ (3) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ (4) $\tan^{-1}\left(\frac{1}{2}\right)$

13) If $\sin^{-1} x + \cot^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$, then x is equal to

(1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{5}}$ (3) $\frac{2}{\sqrt{5}}$ (4) $\frac{\sqrt{3}}{2}$

14) The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

(1) 1 (2) 3 (3) $\sqrt{10}$ (4) $\sqrt{11}$

15) The radius of the circle passing through the point (6,2) two of whose diameters are $x + y = 6$ and $x + 2y = 4$ is

(1) 10 (2) $2\sqrt{5}$ (3) 6 (4) 4

16) If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a}, \vec{c}, \vec{b}]$ is equal to

(1) 2 (2) -1 (3) 1 (4) 0

17) If $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}, \vec{b} = \hat{\imath} + \hat{\jmath}, \vec{c} = \hat{\imath}$ and $(\vec{a} \times \vec{b}) \times \vec{c} = \lambda\vec{a} + \mu\vec{b}$, then the value of $\lambda - \mu$ is

(1) 0 (2) -2 (3) 2 (4) -1

18) If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(1) $\frac{\pi}{2}$ (2) $\frac{3\pi}{4}$ (3) $\frac{\pi}{4}$ (4) π

19) The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{2}$

20) If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$, then

(1) $c = \pm 3$ (2) $c = \pm\sqrt{3}$ (3) $c > 0$ (4) $0 < c < 1$

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II. Answer any 7 of the following .Q.No 30 is compulsory 7x2=14

21. If A is a nonsingular matrix of odd order, prove that $|\text{adj } A|$ is positive.
22. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.
23. If $z = x + iy$, find $Re(i\bar{z})$ in rectangular form.
24. Find a polynomial equation of minimum degree with rational coefficients, having $2i + 3$ as a root.
25. Sketch the graph of $y = \sin\left(\frac{1}{3}x\right)$ for $0 \leq x < 6\pi$.
26. Examine the position of the point (2,3) with respect to the circle $x^2 + y^2 - 6x - 8y + 12 = 0$.
27. If $y = 4x + c$ is a tangent to the circle $x^2 + y^2 = 9$ find c,
28. If $a = \hat{\imath} - 2\hat{\jmath} + 3\hat{k}, \vec{b} = 2\hat{\imath} + \hat{\jmath} - 2\hat{k}, \vec{c} = 3\hat{\imath} + 2\hat{\jmath} + \hat{k}$, find $\vec{a} \cdot (\vec{b} \times \vec{c})$.
29. Find the intercepts cut off by the plane $\vec{r} \cdot (6\hat{\imath} + 4\hat{\jmath} - 3\hat{k}) = 12$ on the coordinate axes.
30. Show that the equation of the circle for which $(-4, -2)$ and $(-1, -1)$ are the ends of a diameter is $x^2 + y^2 + 5x + 3y + 6 = 0$

III. Answer any 7 of the following .Q.No 40 is compulsory 7x3=21

31. If the vectors $2\hat{\imath} - \hat{\jmath} + 3\hat{k}, 3\hat{\imath} + 2\hat{\jmath} + \hat{k}, \hat{\imath} + m\hat{\jmath} + 4\hat{k}$ are coplanar, then find the value of m.
32. Find the rank by row reduction method $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$
33. Determine the values of λ for which the following system of equations $x + y + 3z = 0, 4x + 3y + \lambda z = 0, 2x + y + 2z = 0$ has a unique solution.
34. Simplify $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$ into rectangular form.
35. If $|z| = 2$, show that $8 \leq |z + 6 + 8i| \leq 12$.
36. Show that the equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has atleast 6 imaginary solutions.
37. Find the value of $\tan^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$.
38. Prove that the point of intersection of the tangents at 't_1' and 't_2' on the parabola $y^2 = 4ax$ is $[at_1t_2, a(t_1 + t_2)]$.
39. Find the angle between the planes $\vec{r} \cdot (\hat{\imath} + \hat{\jmath} - 2\hat{k}) = 3$ and $2x - 2y + z = 2$.
40. If $adj(A) = \begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}$, find $|adj(adj(A))|$.

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IV. Answer the following 7x5=35

41. A man is appointed in a job with a monthly salary of certain amount and a fixed amount of annual increment. If his salary was ₹19,800 per month at the end of the first month after 3 years of service and ₹23,400 per month at the end of the first month after 9 years of service, find his starting salary and his annual increment. (Use matrix inversion method to solve the problem.)
(OR) If $z = x + iy$ is a complex number such that $Im\left(\frac{2z+1}{iz+1}\right) = 0$, show that $2x^2 + 2y^2 + x - 2y = 0$.

42. $Solve\ 6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$ (OR) Find the equation of the circle passing through the points $(1,1), (2,-1)$, and $(3,2)$.

43. If $ax^2 + bx + c$ is divided by $x+3, x-5$, and $x-1$, the remainders are 21,61 and 9 respectively. Find a, b and c. (Use Gaussian elimination method.)(OR) Prove by vector method that $sin(\alpha + \beta) = sin\,\alpha\, cos\,\beta + cos\,\alpha\, sin\,\beta$.

44. Show that the points $1, \frac{-1}{2} + i\frac{\sqrt{3}}{2}$, and $\frac{-1}{2} - i\frac{\sqrt{3}}{2}$ are the vertices of an equilateral triangle. (OR) Find the value of $cos\left(sin^{-1}\left(\frac{4}{5}\right) - tan^{-1}\left(\frac{3}{4}\right)\right)$.

45. Solve, by Cramer's rule, the system of equations $x_1 - x_2 = 3, 2x_1 + 3x_2 + 4x_3 = 17, x_2 + 2x_3 = 7$. (OR)
If $\vec{a} = 2\hat{\imath} + 3\hat{\jmath} - \hat{k}, \vec{b} = 3\hat{\imath} + 5\hat{\jmath} + 2\hat{k}, \vec{c} = -\hat{\imath} - 2\hat{\jmath} + 3\hat{k}$, verify that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a}.\vec{c})\vec{b} - (\vec{b}.\vec{c})\vec{a}$

46. Solve : $tan^{-1}\left(\frac{x-1}{x-2}\right) + tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$.(OR) Find the vector parametric, vector non-parametric and Cartesian form of the equation of the plane passing through the points $(-1,2,0), (2,2,-1)$ and parallel to the straight line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$.

47. At a water fountain, water attains a maximum height of 4m at horizontal distance of 0.5m from its origin. If the path of water is a parabola, find the height of water at a horizontal distance of 0.75m from the point of origin. (OR) Find the centre, foci of the hyperbola $11x^2 - 25y^2 - 44x + 50y - 256 = 0$ Also find latus rectum

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